

Use Case 1: Payment and Subscription Management

Field	Details
Use Case Name	Payment and Subscription Management
Use Case ID	UC-10
Author	System Architect
Purpose	To allow subscribers to choose a subscription plan, process payments/renewals, manage billing details, check status, and cancel subscriptions.
Requirement Traceability	BR-10
Priority	High
Preconditions	- Subscriber is logged into the application. - Internet connection is active. - Payment Gateway integration is operational.
Postconditions	- Subscription plan is created, renewed, or canceled. - Payment transaction record is stored. - Digital invoice/receipt is generated and sent to the subscriber.
Actors	Subscriber, Payment Gateway, Admin

Flow of Actions

Basic Flow:

1. The subscriber navigates to the "Subscription & Billing" section in the application.
2. The system displays available subscription plans, payment options, and the subscriber's current status.
3. The subscriber selects a subscription plan (or chooses to update billing details).
4. The system prompts the subscriber to enter or confirm payment details.
5. The subscriber submits the payment request.
6. The system sends the transaction details to the **Payment Gateway** for verification and processing.
7. The Payment Gateway processes the payment and returns a success confirmation.
8. The system updates the subscriber's account status to active/renewed.
9. The system automatically **generates an invoice/receipt** and displays it on screen while sending a copy to the subscriber's email.
10. The Admin can view the updated subscription status and payment log via the admin portal.

Alternative Flows:

- **Payment Declined / Failed:**

1. The Payment Gateway rejects the transaction (e.g., insufficient funds or invalid card details).
2. The system displays: *"Payment failed. Please check your billing information and try again."*
3. The subscription status remains unchanged.

- **Cancel Subscription:**

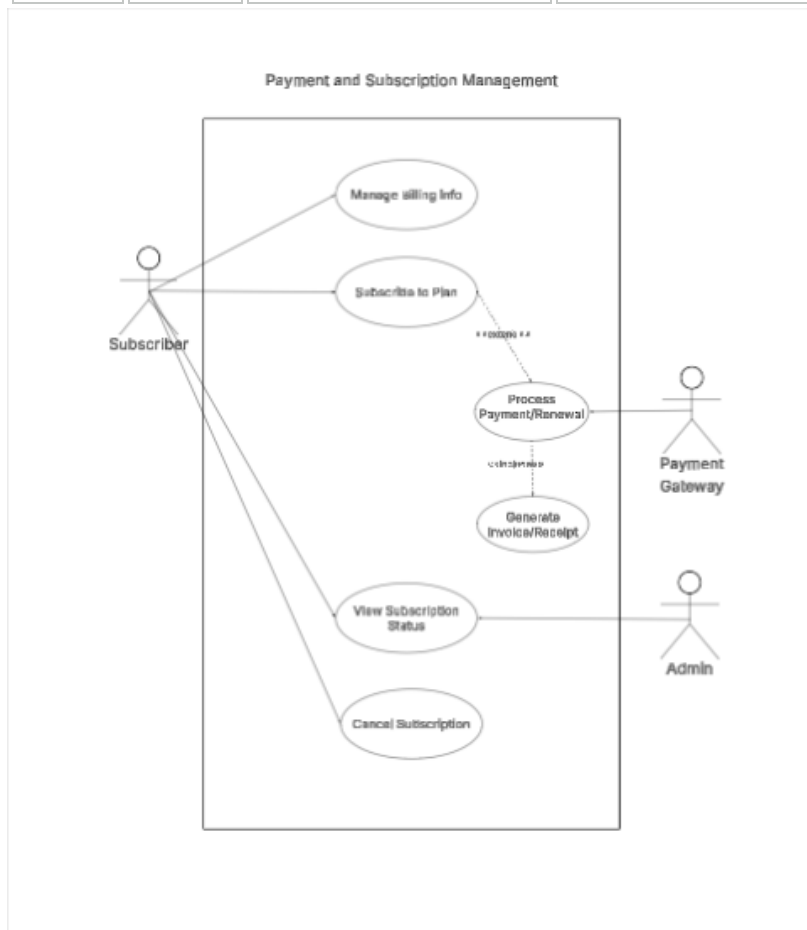
1. The subscriber selects "Cancel Subscription".
2. The system prompts for confirmation and optional feedback.
3. The subscriber confirms cancellation.
4. The system updates the subscription status to set it to expire at the end of the current billing cycle.
5. The system displays: *"Your subscription has been successfully canceled. You will retain access until the end of your billing cycle."*

- **No Internet Connection:**

1. The app detects a network interruption during checkout.
2. The system displays: *"Connection lost. Please check your internet connection and try again."*

Test Cases for UC-10

TC ID	UC ID	Test Case Name	Test Case Description
TC-10	UC-10	Verify Subscription Purchase and Renewal	Ensure that a subscriber can select a plan, process payment via the Payment Gateway, receive a generated invoice/receipt, and successfully activate or extend their subscription status.
TC-11	UC-10	Verify Subscription Cancellation	Ensure that a subscriber can cancel an active subscription and retain access until the current billing cycle expires.



Use Case 2: Premium Content Management

Field	Details
Use Case Name	Premium Content Management
Use Case ID	UC-11
Author	System Architect
Purpose	To allow active subscribers to view exclusive premium articles, download issue/PDF bundles for offline reading, and bookmark content to their personal library.
Requirement Traceability	BR-11
Priority	High
Preconditions	- The user is logged in with an active subscriber account. - Premium content (articles and PDF bundles) is available on the platform. - Internet connection is available.
Postconditions	- Premium content is rendered for the user. - Requested PDF bundle is downloaded to the user's device. - Selected articles are saved to the user's bookmark library.
Actors	Subscriber

Flow of Actions

Basic Flow:

1. The subscriber logs into the system and navigates to the "Premium Content" section.
2. The system verifies active subscription access and displays premium articles and downloadable issue bundles.
3. The subscriber selects a premium article to read.
4. The system loads and renders the full premium article without ad distractions.
5. The subscriber clicks the **"Bookmark"** button on an article.
6. The system saves the article reference to the subscriber's personal library and displays a confirmation: *"Article added to your bookmarks."*
7. The subscriber navigates to the "Issue Bundles" section and taps **"Download Bundle"**.
8. The system package-checks and initiates the PDF bundle download to the subscriber's local device storage.

Alternative Flows:

- **Inactive or Expired Subscription:**
 1. An unauthenticated or free-tier user attempts to open a premium article or download a bundle.
 2. The system intercepts the request and displays: *"This content is exclusive to subscribers. Please upgrade your plan to access."*
 3. The user is provided a prompt to redirect to the Payment and Subscription screen.
- **Content / Download Unavailable:**
 1. The requested bundle file or article is temporarily missing or undergoing updates on the server.
 2. The system displays: *"Content temporarily unavailable. Please try again later."*
- **Network Interruption During Download:**
 1. The connection drops while downloading a PDF bundle.
 2. The app pauses or cancels the download and alerts the user: *"Download failed due to connection error. Tap to retry."*

Test Cases for UC-11

TC ID	UC ID	Test Case Name	Test Case Description
TC-12	UC-11	Verify Access to Premium Articles	Ensure that an active subscriber can view full premium articles, while free/unauthenticated users are prompted to upgrade.
TC-13	UC-11	Verify PDF Bundle Download	Ensure that an active subscriber can download digital magazine PDF bundles to their local device storage for offline reading.
TC-14	UC-11	Verify Article Bookmarking	Ensure that a subscriber can bookmark articles and view saved content within their personal library.

